

Software Requirements Specification (SRS)

1. Introduction

This Software Requirements Specification (SRS) document provides a complete description of the system requirements for the Online Food Delivery Platform. It serves as a baseline for system design, development, testing, and deployment.

2. System Overview

The Online Food Delivery Platform is a web and mobile-based system that enables customers to order food online, restaurants to manage orders digitally, and administrators to monitor and control platform operations.

3. System Architecture Overview

The system architecture consists of a presentation layer (web and mobile applications), an application layer (backend services), a data layer (relational database), and integrations with external services such as payment gateways and notification services.

4. Functional Requirements

SRS ID	System Requirement
SRS-01	The system shall authenticate users using secure login credentials
SRS-02	The system shall encrypt sensitive user data
SRS-03	The system shall retrieve restaurants based on user location
SRS-04	The system shall generate unique order identifiers
SRS-05	The system shall store order status history
SRS-06	The system shall process online payments through integrated gateways

5. Non-Functional Requirements

Category	Requirement
Performance	System response time shall be less than 3 seconds
Scalability	System shall support at least 10,000 concurrent users
Availability	System uptime shall be 99.5%
Security	All data transmission shall use HTTPS

6. Data Requirements

Entity	Key Attributes
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User	User ID, Name, Contact Details
Restaurant	Restaurant ID, Location, Menu
Order	Order ID, User ID, Status, Timestamp
Payment	Transaction ID, Order ID, Payment Status

7. Interface Requirements

Interface Type	Description
User Interface	Web and mobile application interfaces
External Interface	Payment gateway and notification APIs
Internal Interface	Frontend to backend communication

8. Security Requirements

Security Aspect	Requirement
Authentication	Role-based access control
Data Protection	Sensitive data encryption
Compliance	Adherence to data protection regulations

9. Error Handling and Logging

Scenario	System Action
System error	Log error and notify administrator
Payment failure	Rollback transaction and notify user
API timeout	Retry and log incident

10. Constraints and Assumptions

Type	Description
Constraint	Dependent on third-party APIs
Constraint	Fixed project timeline
Assumption	Users have compatible devices

11. Acceptance Criteria

Feature	Acceptance Condition
User Login	Successful login with valid credentials
Order Placement	Order created with valid ID
Payment	Correct payment status update

12. Traceability Matrix

FRD ID	SRS ID
FR-01	SRS-01
FR-05	SRS-03
FR-11	SRS-04
FR-14	SRS-06